

AeroRoute

Process, assumptions, and scenario coverage

Sequential flight-choice environment and reliability verifier

5 October 2026

1. Purpose

The task was to turn "get from one U.S. airport to another" into a sequential decision problem, and to score a finished trip with an explicit reliability rubric. The traveler stands at an airport, chooses a flight from the flights that are scheduled and boardable, and then lives with what comparable historical flights actually did. The episode continues from wherever that outcome leaves the traveler.

A trained policy was not required. The delivered baseline is an earliest-arrival rule: prefer a direct flight to the requested destination, then the earliest scheduled arrival. It does not look at historical outcomes when it chooses. The agent runs that baseline through the Gymnasium adapter, and the verifier checks the resulting trace, both on the published 2025 schedule.

2. Source data

The source is the Bureau of Transportation Statistics Marketing Carrier On-Time Performance table, January 2020 through December 2025. It is the marketing-carrier table, not the reporting-carrier table. All 72 monthly ZIP archives were downloaded and checked: 40,910,253 records and 2,163,774,074 compressed bytes. The normalized snapshot keeps every record, including cancellations and diversions.

Normalization counted 917,084 cancellations and 97,790 diversions. Years 2020 through 2024 are the fitting years. Year 2025 is held out. The 2025 schedule index contains 7,488,673 flights. Raw archives and processed files stay under data/ and are not in Git.

3. Process

Each stage writes an immutable directory named by the digest of its inputs. A later stage opens the earlier directory by that name.

- Download the 72 monthly archives. A rerun skips a file whose checksum already matches. Incomplete months resume. Conflicting completed files are left in place.
- Normalize each month into Parquet. Cancellations, diversions, duplicate flags, and unresolved rows stay in the file. Bad numeric cells become null plus a diagnostic. Timestamps are reconstructed with pinned timezone data, not the host timezone.
- Build a row-group lookup so a later read can fetch one flight without scanning a month.
- Resolve dated operating services. The published catalog contains 39,673,798 services. A service identity is the operating carrier, flight number, origin, destination, and scheduled departure on a given date.

- Index every 2025 departure for schedule search. The index does not drop a flight because of its historical outcome or because it arrives after a deadline.
- Build an exact-replay map from each 2025 service to its own source row, and a pool catalog from 2020-2024 services that have a clean schedule and clean evidence.
- Freeze a scenario as a pilot: one seed, one scenario id, and one example flight. Run the earliest-arrival agent and store the episode, the experiment settings, and the verifier report.



Figure 1. From the monthly archives to a verified episode. Steps 1 through 7 are saved artifacts. Step 8 is one run.

The environment step is then small. It reads departure pages until the offer set is complete, then one historical outcome. It does not scan the 40.9 million source rows.

4. Assumptions

These are modeling choices. They are stored on the episode. They are not quantities observed by BTS.

- `start_at` is the time the traveler is present at the origin. The default initial boarding buffer is 60 minutes, so the traveler can board a flight that departs at `start_at` plus 60 minutes. After a completed flight, the connection buffer is 45 minutes. After a cancellation, the extra boarding buffer is 0.
- Empirical mode uses `whole-timeline-recovery-v1`. Every resolved historical timestamp is shifted by the gap between the selected flight's scheduled departure and the historical flight's scheduled departure. The gaps between events stay as observed. A flight that left 20 minutes after its own

scheduled departure still leaves 20 minutes after the selected departure. Lateness against the selected flight's scheduled arrival is not preserved, because that flight can have a different scheduled duration.

- Replay mode does not shift. The 2025 flight is scored with its own recorded times, including an incomplete record.
- A cancellation is treated as known at the scheduled departure plus 60 minutes. The traveler may then choose another flight. If that historical cancellation also has a departure time, the episode stops as unresolved. A return-to-gate time is not invented.
- If the historical departure is before the traveler is ready, the outcome is missed boarding, observed at the scheduled departure. The missed-boarding buffer is 0. That service cannot be selected again. The initial and connection buffers are not applied a second time.
- A diversion whose gate arrival cannot be timed stops the episode. Wheels-on and wheels-off are not converted into a gate arrival by adding taxi or deplaning time. Div1Airport is not assumed to be the final stop.
- The arrival deadline grades the trip. It does not end the episode. The episode ends when the traveler reaches the requested destination, when the clock reaches start_at plus 24 hours, or after 8 decisions. Arrival exactly at the cutoff counts. A new choice at the cutoff does not.
- The reward is 0 on every nonterminal step. The terminal score is reliability-rubric-v1: destination reached 60, deadline met 25, no cancellation, diversion, or missed boarding 10, and elapsed time known and inside the episode limit 5. Each part scores 0 or 1. The reward is the weighted sum divided by 100. A clean on-time arrival scores 1.0.
- The trace does not store a success flag. The verifier rebuilds the offer set, redraws the same outcome, reapplies the same transition rules, and recomputes the score.
- A pool is chosen by relaxing the match in this order: same route, carrier, quarter, weekday, and 6-hour local departure bucket; then drop weekday; then drop the time bucket; then drop the quarter, leaving carrier and route; then route only. A level is used when it has at least 30 historical services, except the route-only level, which is used even when support is lower. Each service in the chosen pool is equally likely. Extra source rows for the same service do not add weight.
- The draw is a function of the scenario id, the seed, the selected service, and the frozen pool membership. The same pilot returns the same historical flight on every reset. The Gymnasium reset seed affects only adapter randomness.

5. Environment design

The environment is a single traveler moving through one itinerary. A step is one flight choice. The state the policy needs is where the traveler is, when a decision can be made, when boarding is allowed, and which flights are still legal. Future outcomes stay with the provider. They are not part of the observation.

- State. Current airport, decision time, ready-to-board time, the trip origin, destination, and deadline, decisions already taken, and the services already attempted. After a normal arrival, the next

decision time is the gate arrival, and ready-to-board is that time plus the 45-minute connection buffer. After a cancellation, only the cancellation boarding buffer applies. After a missed boarding, only the missed-boarding buffer applies, which is 0.

- Action. One service from the flights whose scheduled departure is at or after ready-to-board and before the episode cutoff. A service already attempted is excluded, including its aliases. An action that was not offered does not change the episode.
- Observation, for the Gymnasium adapter. flights is a padded array of origin id, destination id, scheduled departure and arrival in minutes after start_at, and a flag for whether that leg ends at the trip destination. state holds the current airport, the trip destination, decision and ready minutes, the deadline, the episode length, the decision count, and a done flag. time_known marks whether decision time and ready time are known. action_mask marks the legal slots. The last slot is only legal when the episode is already over and must be acknowledged.
- Transition. The provider returns one historical outcome. Empirical mode shifts that timeline onto the selected departure. Replay keeps the 2025 times. The traveler then arrives at the scheduled destination, stays for a cancellation or missed boarding, continues from a diversion whose gate time is known, or stops when the outcome cannot be timed.
- Stopping. The episode ends on arrival at the requested destination, at start_at plus 24 hours, or after 8 decisions. The deadline is recorded for the score. It does not end the episode. Reward on every earlier step is 0.

6. Verifier formula

The verifier does not read a stored success flag or a stored reward. It checks the trace, rebuilds the offer set from the schedule, redraws the same historical outcome, reapplies the same transition rules, and scores the finished trip with reliability-rubric-v1. Each criterion is 1 or 0.

$$R = (60 d + 25 t + 10 c + 5 e) / 100$$

Symbol	Criterion	Score is 1 when
d	Destination	Arrival time at the requested airport is known
t	Deadline	That arrival is at or before the deadline
c	No disruption	No cancellation, diversion, or missed boarding
e	Known elapsed time	The trace end is known and within the limit

A clean on-time arrival is $d = t = c = e = 1$, so $R = 1$. A trip that arrives late but without a disruption is $R = 0.75$. A trip that reaches the destination on time after a cancellation is $R = 0.90$. A trip that never arrives, misses the deadline, and has a disruption, but ends on a known clock inside the limit, is $R = 0.05$. If the ending time itself is unknown, e is 0 as well.

7. What was kept, and what sampling skips

The normalized files were not trimmed to the flights that are easy to score. Cancellations and diversions remain. Rows with an airport-reference disagreement remain: 2,462,259 flight rows are flagged, mostly because the airport reference labels DCA and IAD as the District of Columbia while BTS says Virginia, and because some territory labels differ. Those timezones were not guessed and the rows were not dropped.

Two smaller sets never become an offer the agent can select:

- 1,236,454 source rows have a scheduled departure or arrival that could not be resolved to a timestamp. They stay in the normalized month. They do not receive a service identity, because a bookable flight needs both scheduled times.
- One source row has no flight number: Frontier F9 on 25 August 2024, Miami to Atlanta, scheduled 06:00 to 08:07, source ordinal 423052. Operating and marketing flight numbers are blank. It is a diagnostic, not a service.

Those two exclusions account for the gap between 40,910,253 source rows and 39,673,798 services. Every other source row received a service identity.

A further 98 services stay in the catalog and stay out of the historical pools. They are 49 April 2020 Spirit alias groups. In each group, two services share the date, origin, carrier, flight number, and scheduled departure, and differ only in destination. One pair is NK 1 on 7 April 2020 from Fort Lauderdale at 21:52, one copy to Chicago O'Hare and one to Orlando. The catalog records the collision. Pool membership requires an empty schedule issue and an empty evidence issue, under the rule omit-unresolved-services-v1. The historical catalog is therefore not marked ready. The 2025 schedule has no such issues. The 98 services and the Frontier row were not deleted.

Of the 7,488,673 flights on the 2025 schedule, 7,468,740 have a historical pool and 19,933 (0.266 percent) do not. A request for one of those 19,933 raises. The environment does not silently draw a different flight and does not claim that the schedule had no flights.

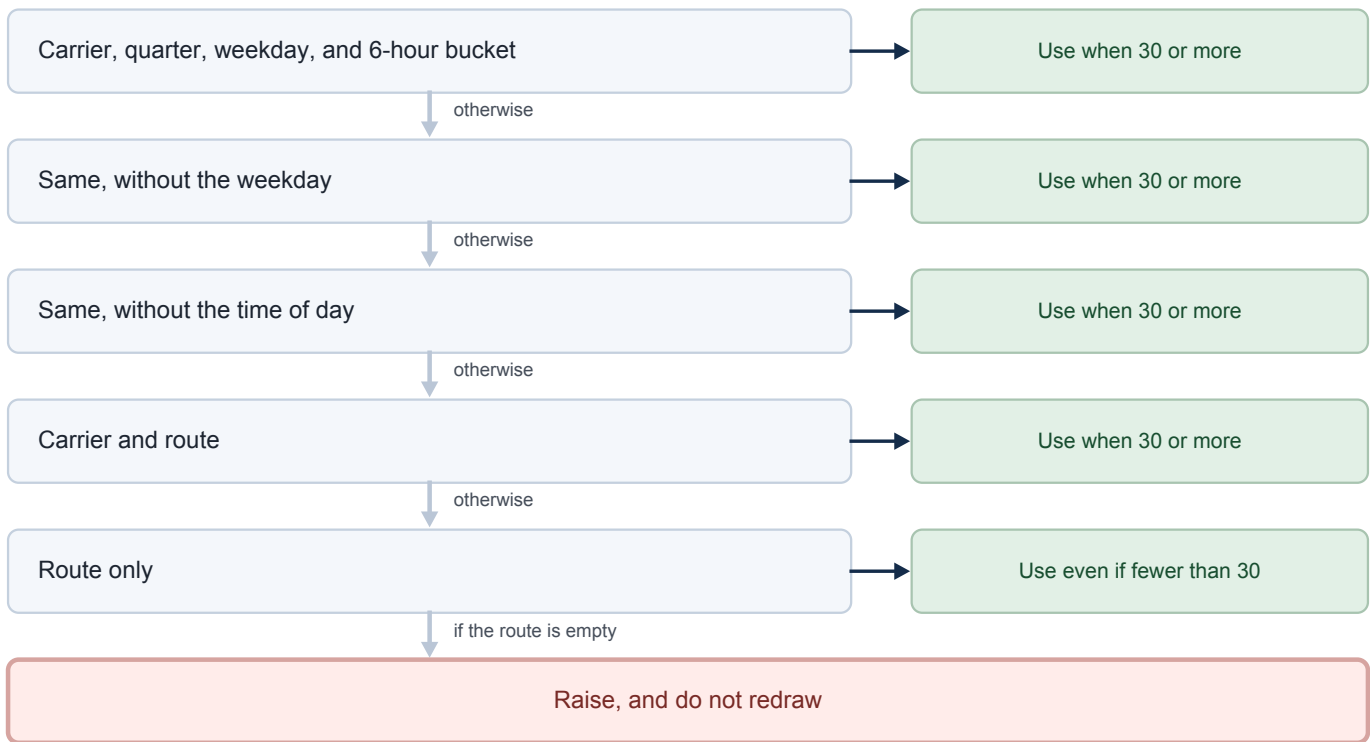


Figure 2. Pool choice for one 2025 flight. The first level with enough services wins. Each service in that pool is equally likely.

Shares below are of all 7,488,673 scheduled 2025 flights, including the 19,933 with no pool:

Match	Flights	Share
Carrier, quarter, weekday, 6-hour bucket	5,405,754	72.2%
Carrier, quarter, 6-hour bucket	1,562,384	20.9%
Carrier and quarter	299,637	4.0%
Carrier and route	116,316	1.55%
Route only	84,649	1.13%

500,602 flights lose the time-of-day criterion on the way down that list. 7,534 flights sit in a pool smaller than the 30-service target; those pools are still used because the route-only level is the last resort. 84,649 flights fall all the way to route only, so the draw may come from a different carrier and a different time of day.

8. Scenarios that run

The earliest-arrival agent was run through the Gymnasium adapter on frozen empirical scenarios. The verifier accepted each trace. The index of pilots and episodes is `data/processed/access/pilots/review_cases.json`. Two on-time rows both use seed 0 because they are different scenarios, train-scenario and review-on-time. A seed fixes the draw only together with its scenario id.

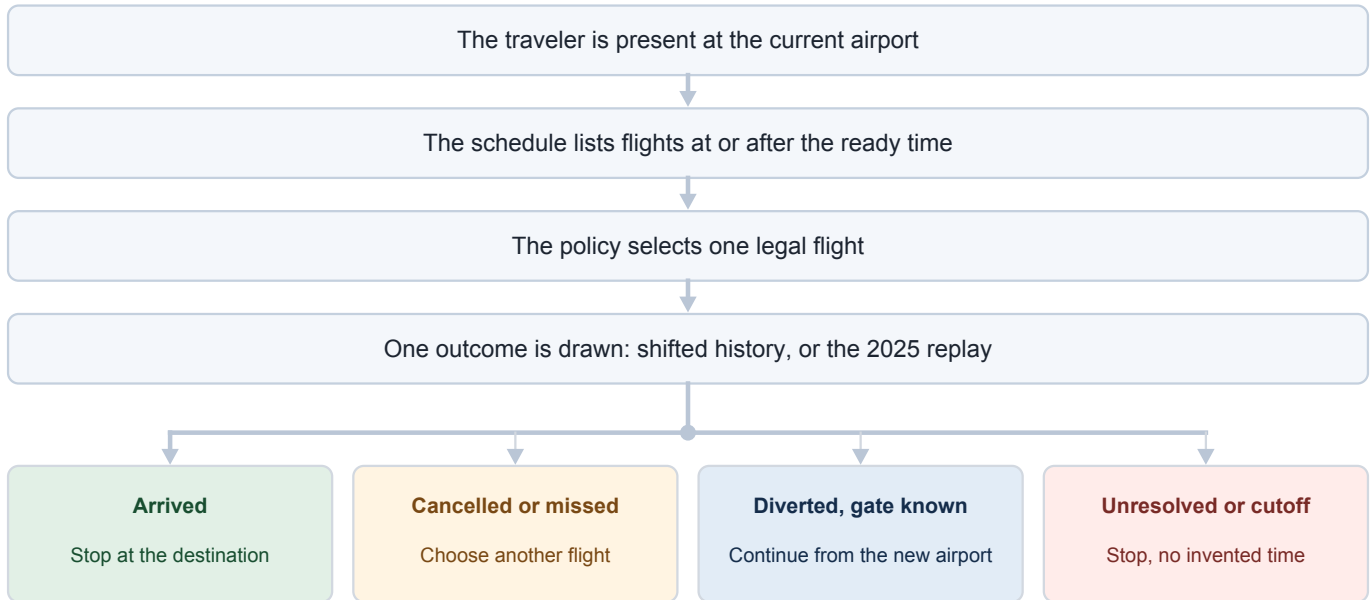


Figure 3. One episode. The deadline grades the trip. Arrival, the time limit, or the decision limit is what stops it.

Outcome	Route	Reward	What happened
On time (3)	MCI-TPA	1.00	One flight, deadline met
Cancellation (2)	MCI-TPA	0.05	Cancel, then unresolved
Cancellation (1)	BNA-ATL	0.90	Cancel, then a later arrival
Missed boarding (3)	MCI-TPA	0.05	Left early, then unresolved
Late arrival (3)	BIL-SLC	0.75	Arrived after the deadline
Diversion (1)	MIA-LAX	0.90	Diverted, deadline still met
Diversion (1)	CLT-IAH	0.65	Diverted, arrived late

The scores follow the rubric directly. A cancellation or missed boarding removes the 10-point disruption component. Missing the destination removes 60. Missing the deadline removes 25. An episode that reaches the time limit with the clock known still earns the 5-point elapsed-time component, which is the 0.05 score. Nashville to Atlanta shows the other cancellation shape: the agent recovers onto a later flight, reaches Atlanta by the deadline, and loses only the disruption component, scoring 0.90.

Replay is the other mode that runs. It returns the held-out 2025 row for the flight the traveler selected, including a cancellation, a diversion, or an incomplete record, and it does not fall back to a 2020-2024 pool. The same Kansas City to Tampa request was run in both modes before these review pilots were frozen.

9. Scenarios that stop

These are limits of the published scenarios, not failures of the source archive.

- The historical catalog is not ready. The 49 Spirit alias groups and the blank Frontier identity are still open. Sampling skips them. It does not decide which of the two Spirit destinations was the real flight.
- A 2025 itinerary scored with a 2020-2024 draw is a training distribution. It is not a measurement of how those 2025 flights actually performed. That measurement is replay.
- Most diverted historical flights in the pools examined here had already departed before the traveler was ready. The environment records missed boarding and does not also record the diversion. Only two review draws stayed diversions: Miami to Los Angeles and Charlotte to Houston.
- A stranded diversion whose gate time cannot be reconstructed does not become an arrival. Across the pools checked for the review set, the flight the baseline would actually pick usually did not contain that stranded flight. The search was stopped once each outcome had two or three examples.
- A cancellation that also reports a departure time ends unresolved. No gate-return timeline is created.
- 19,933 scheduled 2025 flights have no pool. Selecting one is an error, not a completed episode with reward 0.
- A scheduled departure of 2400 always stays unresolved. An observed 2400 clock can be read as midnight only when another timestamp already supplies the date.
- Airport-reference conflicts were preserved. A downstream timezone that depended on a guessed label was not produced.

10. Future work

- Review the 49 Spirit alias groups and the 25 August 2024 Frontier row, and record the decision in a versioned review layer. The historical catalog can be marked ready only after that review. The rows should stay in the source either way.
- Reconcile the airport and timezone reference disagreements, especially DCA, IAD, and the territory labels, with an explicit override list rather than a silent repair.
- Decide how a real 2025 evaluation uses held-out outcomes. Training can keep drawing from 2020-2024. Reporting 2025 performance should say whether the score is replay, a sealed empirical scenario, or something else.
- Decide whether years inside a pool should be weighted. The current pools treat every fitting-year service equally.
- Train policies against the Gymnasium adapter and compare them with earliest-arrival and masked random on the same frozen pilots. Reset seeds do not create new historical draws. New draws need new scenarios.
- Add a multi-request evaluation harness. The review set is a handful of hand-chosen routes, not a sample of U.S. itineraries.
- Publish pool membership beside the pool definitions so the first use of a route does not recount the

historical services.

- Write the alias and identity conflicts into a small report tied to the catalog checksum, so the next build does not rediscover them with another full scan.